A Meta-Model-Guided Symbolic Reasoning Method for Solving Trigonometric Function Problems

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***Abstract*—Automated trigonometric function problem (TFP) solving is a pivotal yet challenge task in intelligent education systems. Unlike general algebraic problems, trigonometric reasoning involves rigorous angle and interval constraints, specialized transformations, and infinite periodic solution structures. Although existing studies have explored the proof and reduction of trigonometric expressions, they lack a systematic investigation into trigonometric properties and fail to establish algorithms for TFP solving. In this paper, we propose a Meta-Model-Guided Symbolic Reasoning Framework to solve trigonometric function problems. We formalize representative trigonometric reasoning operations into structured trigonometric meta-models (TMMs), which execute symbolic actions while strictly preserving trigonometric properties and constraints. We then develop a decoupled reasoning strategy (DRS) that dynamically selects and executes TMMs conditioned on the problem constraints and solving goals. Evaluations on a benchmark of 795 trigonometric function problems demonstrate that our framework achieves an accuracy of 74.8%, significantly outperforming large language model baselines. The proposed framework provides a robust solution for advancing intricate and specialized mathematical reasoning.**

***Index Terms*—trigonometric function problem solving, symbolic reasoning, trigonometric meta-model, decoupled reasoning strategy, intelligent education**

I. Introduction

Automatic mathematical problem solving is a foundational technology for intelligent education systems, requiring solvers to produce both exact answers and tutorable derivation steps [1], [2]. Among various mathematical subjects, trigonometric function problem (TFP) solving represents a critical yet demanding task. As an essential branch of mathematics, trigonometric functions involve distinct computational operations and specialized reasoning rules. Consequently, such solving algorithms serve as foundational technologies for advancing reasoning methods and intelligent education systems. However, existing research on automated mathematical problem solving has predominantly focused on general arithmetic and algebraic problems. This limitation underscores the need to establish solving methods for trigonometric function problems.

Unlike general algebraic problems, trigonometric function problems involve unique mathematical properties and constraints. First, trigonometric functions are strictly restricted by angle conventions and domain constraints, such as quadrant ranges and interval boundaries. Second, trigonometric function problem solving relies on distinctive identity transformations, which must preserve mathematical equivalence under variable and equation derivation. In addition, trigonometric equations and inequalities naturally yield multiple solution branches that require exhaustive consideration. Furthermore, the inherent periodicity of trigonometric functions often requires extending local results into infinite periodic solution sets over the real domain. These characteristics are closely connected during problem solving. Missing any constraint or solution branch can lead to an incorrect final answer.

Existing studies are limited to either elementary function problem solving or subtasks of trigonometric reasoning. In the broader domain of funtion problem solving, some algorithms have been developed to solve six types of elementary function, including linear, quadratic, inverse proportional, exponential, logarithmic, and power functions [1], [3]. However, these methods are designed for algebraic functions and cannot handle the specialized transformations and periodic structures of trigonometric functions. Meanwhile, research on trigonometry has mainly focused on certain subtasks, such as automated identity proving and expression reduction [4], [5]. In addition, recent benchmarks have evaluated the performance of large language models (LLMs) on trigonometric expression simplification [6]. Therefore, a systematic framework and algorithm for trigonometric function problem solving remain largely unexplored.

To address these challenges, this paper presents a meta-model-guided symbolic reasoning framework for solving trigonometric function problems. Following the established two-phase problem solving paradigm [1], our framework first converts the raw problem narrative into an understood state via constrained semantic parsing [7]. For the core reasoning process, we formalize representative trigonometric operations into nine structured trigonometric meta-models (TMMs). Each TMM encapsulates an atomic symbolic action designed to preserve mathematical equivalence, interval boundaries, and angle conditions during intermediate state transitions. Building upon these models, we develop a decoupled reasoning strategy (DRS) to coordinate multi-step derivations. Conditioned on the problem constraints, DRS dynamically schedules and executes applicable TMMs to satisfy the target solving goals, such as trigonometric equation solving and functional property derivation. Experiments on a benchmark of 795 trigonometric function problems show that our framework achieves a solving accuracy of 74.8%, markedly outperforming LLM baselines. These results demonstrate that the proposed framework offers a reliable and scalable method for enhancing complex and domain-specific mathematical reasoning.

The primary contributions of this paper are as follows:

1) A meta-model-guided symbolic reasoning framework is established to solve trigonometric function problems.
2) A set of trigonometric meta-models (TMMs) is formalized to encapsulate specialized trigonometric operations while preserving properties and constraints.
3) A decoupled reasoning strategy (DRS) is developed to dynamically schedule and execute TMMs.

II. Related Work

A. Trigonometric Symbolic Reasoning

Research dedicated to trigonometric reasoning has mainly focused on restricted formal tasks. AutoTrig formulates identity proving as a search over normalized trigonometric expressions and transformation rules, demonstrating the value of a bounded action space for generating short proofs [4]. TRIGO evaluates generative models on trigonometric proof reduction in Lean and emphasizes step-level formal verification [6]. SSC-CoT and TriMaster100 instead investigate stepwise consistency for language-model reasoning over complex trigonometric problems [5]. These studies provide useful mechanisms for rule search, formal checking, and process-level evaluation. However, identity proving or expression reduction does not cover the problem-level semantics required for function properties, interval constraints, equations, inequalities, and complete periodic solution sets. TrigSolver extends beyond a single formal operation by representing these structures in a shared state and coordinating their execution through goal-conditioned inference.

B. Language Models and Computer Algebra for Mathematical Solving

Large language models and multimodal models are increasingly evaluated on broad mathematical benchmarks such as MATH, MathVista, MathVerse, CMM-Math, and UGMath-Bench [8]–[12]. Their generality is useful for interpreting heterogeneous problem statements, but aggregate benchmark performance does not establish reliable handling of trigonometric equivalence, singularities, or infinite solution families. Chain-of-thought and self-consistency methods improve generated reasoning at the sequence level, although they do not intrinsically enforce symbolic validity or set completeness.

Computer algebra systems complement language models with exact operations such as trigonometric simplification, equation solving, continuous-domain analysis, and symbolic set construction [13]. Nevertheless, a computer algebra result is only as appropriate as the formal query supplied to it. A generic CAS does not decide which problem constraints are implicit, whether a one-period result must be lifted to all real solutions, or whether a transformed expression remains equivalent on the original domain. Neuro-symbolic pipelines that connect language models and symbolic tools partially address calculation reliability, but without a task-specific intermediate state and an explicit inference controller, they remain vulnerable to incorrect tool selection and incomplete answers. TrigSolver uses a language model only for grounded semantic mapping and places mathematical decision making in the Trig-URM, TMM, and validation layers.

C. Structured Function Problem Solving

Relation-centric approaches to text-diagram function problems represent extracted mathematical relations before symbolic solving, thereby separating problem understanding from equation execution [1]. Subsequent work on uniform representation and decoupled inference further suggests that diverse function types can share a procedural layer while retaining type-specific meta-models [3]. Trigonometric functions, however, introduce structures that are not adequately captured by representations designed for elementary algebraic functions. Their values and solution sets depend on angle normalization, principal branches, periodic equivalence, singular points, and repeated interval units. This work specializes and extends the structured function-solving paradigm through a periodicity-aware representation and a TMM library whose transitions explicitly construct and verify complete periodic sets.

III. Problem Formulation

A. Problem Statement

Let a trigonometric function problem be represented as

$$P = (T, \mathcal{O}), \tag{1}$$

where T denotes the natural-language problem text containing mathematical expressions and conditions, and $\mathcal{O}$ represents an optional set of candidate options for multiple-choice questions.

The primary objective of trigonometric function problem solving is to derive a solution from the input P. We formalize the solver output as a structured tuple:

$$S = (A, \mathcal{E}), \tag{2}$$

where A denotes the primary answer, such as an exact value, an algebraic expression, or a functional property. The term $\mathcal{E}$ represents the exact solution set, which can be an finite subset or an infinite periodic set over the real domain $\mathbb{R}$.

Similar to many existing algorithms, the proposed algorithm comprises two phases of problem understanding and symbolic solver. The intermediate results produced by a solving algorithm are also called problem states. The understood state defined below is one of the critical states.

Definition 1 (*Understood State*). An intermediate representation $U^{(0)}$ is called an understood state of problem P if it encapsulates all explicit mathematical entities, angle conditions, domain constraints, and solving goals required for the symbolic solver to derive the final solution S without revisiting the original text T.

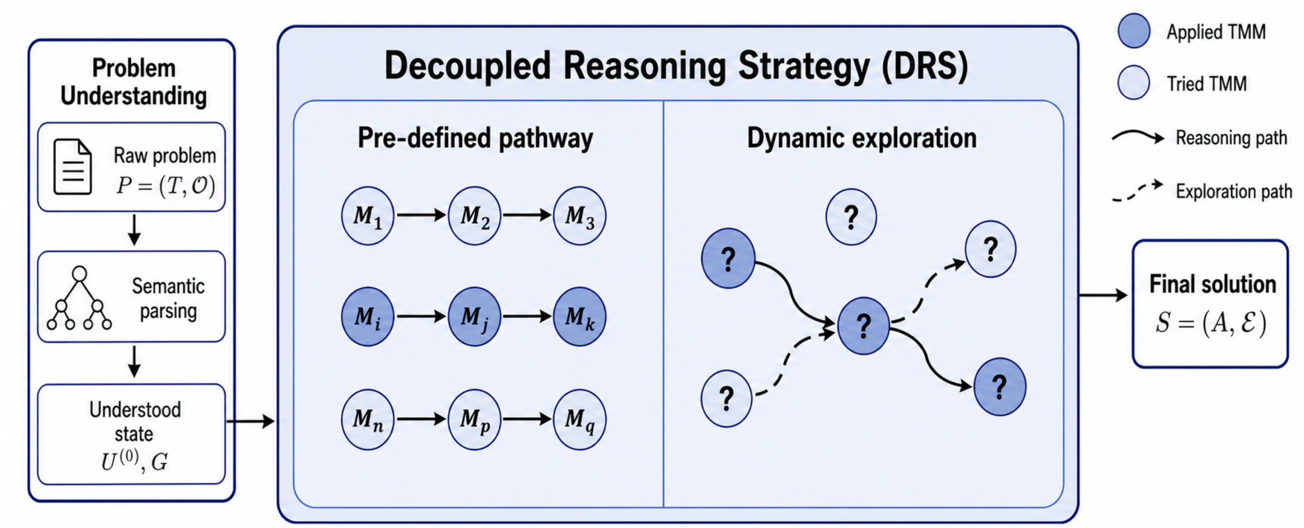

Fig. 1: Overview of the proposed framework for trigonometric function problem solving. Decoupled reasoning strategy coordinates the trigonometric meta-models through pre-defined pathways and dynamic exploration to derive solutions from understood state.

The understood state $U^{(0)}$ serves as a standardized interface between text understanding and symbolic reasoning. In the problem understanding phase, semantic parsing extracts the stated mathematical formulas, angle ranges, and target objectives from T. Once $U^{(0)}$ is established, the subsequent symbolic solver operates entirely on this representation.

B. Research Scope

This work focuses on trigonometric function problems that require explicit trigonometric reasoning. These problems examine the formal understanding of trigonometric concepts, algebraic identities, function properties, and exact symbolic expressions.

Accordingly, several adjacent mathematical tasks fall outside the scope of this study. Specifically, multimodal trigonometric function problems, geometric problems dominated by triangle calculations, and some problems containing non-elementary transcendental equations are not addressed in the current framework.

IV. Meta-Model-Guided Symbolic Reasoning

A. Architecture Overview

Deriving the solution for a trigonometric function problem from an understood state requires coordinating heterogeneous mathematical operations. Unlike elementary algebraic problems where operations can follow fixed procedural routines, trigonometric problem solving presents a dual challenge: local symbolic transformations must strictly preserve domain and angle validity, while the overall search trajectory must adapt dynamically to diverse solving goals. To address these challenges, we propose a two-layer symbolic reasoning framework, as illustrated in Fig. 1.

As shown in Fig. 1, the framework decouples the reasoning process into two cooperative layers:

- *Trigonometric Meta-Model (TMM) layer*: encapsulates specialized trigonometric operations into self-contained symbolic units, ensuring that each atomic step maintains mathematical equivalence and domain constraints.
- *Decoupled Reasoning Strategy (DRS) layer*: evaluates intermediate problem states and dynamically selects applicable TMMs to produce the final answer.

The reasoning process begins from the initial understood state $U^{(0)}$. The DRS layer iteratively selects and executes matching TMMs to advance intermediate problem states. This deductive process continues until the solving goal is fully satisfied, ultimately producing the final solution S.

B. Trigonometric Meta-Models (TMMs)

Most trigonometric transformation rules typically alter expression syntax without examining prerequisite mathematical conditions, such as non-zero denominators or quadrant restrictions. Unconditioned execution can easily introduce irrelevant solutions or violate domain constraints. To ensure that every symbolic step is mathematically valid and verifiable, we encapsulate typical trigonometric operations into structured units termed trigonometric meta-models.

Definition 2 (*Trigonometric Meta-Model*). A trigonometric meta-models is formalized as a five-tuple:

$$\mathcal{M} = \langle I, \mathcal{P}, \mathcal{O}, \mathcal{T}, \mathcal{V} \rangle, \tag{3}$$

where:

- I denotes a unique model identifier;
- $\mathcal{P}(U)$ represents an applicability precondition specifying the structural patterns, angle conventions, and domain constraints required on state U;
- $\mathcal{O}$ represents a symbolic operation that executes domain-specific algebraic computations;

Algorithm I: Decoupled Reasoning Strategy (DRS)

Input: Understood state $U^{(0)}$, Solving goal G, Pathway library $\mathbb{K}$, TMM library Λ

Output: Final solution tuple $S = (A, \mathcal{E}, \mathcal{T})$

I-1: Reasoning with Pre-defined Pathway;
▷ **if** $(U^{(0)}, G)$ *matches a standard pattern in* $\mathbb{K}$ **then**
  Retrieve canonical pathway $\pi = [\mathcal{M}_1, \mathcal{M}_2, \ldots, \mathcal{M}_k] \leftarrow \mathbb{K}(U^{(0)}, G)$;
  $U \leftarrow U^{(0)}$;
  foreach $\mathcal{M}_i \in \pi$ **do**
    $U \leftarrow \mathcal{T}(\mathcal{M}_i, U)$;
I-2: Reasoning with Exploratory Pathway;
▷ **else**
  $U \leftarrow U^{(0)}$;
  while U *updates* ***and*** $\neg\,\mathrm{GoalSatisfied}(U, G)$ **do**
    $U_{\text{prev}} \leftarrow U$;
    foreach $\mathcal{M}_j = \langle I, \mathcal{P}, \mathcal{O}, \mathcal{T}, \mathcal{V} \rangle \in \Lambda$ **do**
      if $\mathcal{P}_j(U) =$ *true* **then**
        $U \leftarrow \mathcal{T}(\mathcal{M}_j, U)$;
    if $U = U_{prev}$ **then**
      break;
$S \leftarrow \mathrm{ExtractSolution}(U, G)$;
return S;

- $\mathcal{T}(U, e)$ denotes a state transition function that integrates the operational output e into state U;
- $\mathcal{V}(U, U')$ specifies a local validation condition that verifies mathematical equivalence and constraint consistency between state U and candidate state U'.

Given an intermediate state $U^{(j)}$, the execution of model $\mathcal{M}$ is governed by a transition mechanism:

$$U^{(j+1)} = \begin{cases} \widetilde{U}, & \text{if } \mathcal{P}(U^{(j)}) \wedge \mathcal{V}(U^{(j)}, \widetilde{U}), \\ \emptyset, & \text{otherwise}, \end{cases} \tag{4}$$

where $\widetilde{U} = \mathcal{T}(U^{(j)}, \mathcal{O}(U^{(j)}))$ denotes the candidate successor state.

As summarized in Table I, we construct a library Λ containing nine core TMMs organized into four cross-task operation groups. This modular formulation allows individual TMMs to be flexibly reused across different problem categories.

C. Decoupled Reasoning Strategy (DRS)

While TMMs guarantee the correctness of single-step operations, solving a complete trigonometric problem requires coordinating multiple models in sequence. Fixed solving procedures cannot easily accommodate the diverse conditions and varied transformations across different problems. To address this challenge, we introduce the decoupled reasoning strategy (DRS) to dynamically guide the solving process based on intermediate problem states.

The DRS manages the inference process through two complementary mechanisms: pre-defined pathways for standard problem patterns and dynamic exploration for composite problems.

For common trigonometric function problems, the solving process typically follows established mathematical procedures. These problems encompass five foundational categories, respectively querying exact numerical values, simplified algebraic identities, analytical functional properties, complete equation solution sets, and inequality intervals. Based on standard curriculum requirements, we construct five canonical reasoning pathways corresponding to these solving goals:

- *Exact evaluation*: `AngleNormalize` → `ExactEvaluate` → `AnswerValidate`.
- *Identity transformation*: `AngleNormalize` → `IdentityRewrite` → `AnswerValidate`.
- *Property derivation*: `AngleNormalize` → `SinusoidCanonicalize` → `PropertyDerive` → `AnswerValidate`.
- *Equation solving*: `AngleNormalize` → `EquationBaseSolve` → `PeriodicComplete` → `AnswerValidate`.
- *Inequality reasoning*: `AngleNormalize` → `DomainRangeInequality` → `PeriodicComplete` → `AnswerValidate`.

When an understood state and its solving goal match one of these standard patterns, DRS directly activates the corresponding pathway, ensuring efficient and deterministic solving.

When a problem involves composite conditions or solving goals that do not match any pre-defined pathway, DRS activates dynamic exploratory reasoning. In this mode, DRS iteratively inspects the preconditions of all available TMMs in library Λ against the current problem state. Each applicable model is executed, and its validated output updates the problem state. This iterative process continues until the solving goal is satisfied or the state reaches convergence, ensuring that all derivable mathematical knowledge is systematically extracted.

V. Experiments

A. Dataset

To rigorously evaluate symbolic reasoning capabilities, we construct a dataset of 795 trigonometric function problems derived from CMM-Math dataset [11]. We enforce strict scope-filtering criteria: (1) retaining text-based, single-target problems requiring exact algebraic or periodic solutions; and (2) filtering out image-dependent geometry questions, multi-part sub-questions, and non-elementary transcendental systems.

B. Baselines

We evaluate our framework against two mainstream state-of-the-art large language models: Qwen3.5-Flash and DeepSeek-V3. All methods are evaluated directly on the benchmark of 795 trigonometric function problems under identical input statements to measure their end-to-end solving accuracy.

TABLE I: The Pool of trigonometric meta-model in Λ.

Model Identifier (I)	**Precondition** ($\mathcal{P}$)	**Operation** ($\mathcal{O}$) **& State Effect** ($\mathcal{T}$)
`AngleNormalize`	Non-standard angle units or stated quadrant conditions	Converts angles into standard radians and records active quadrant boundaries.
`ExactEvaluate`	Special angle values or known trigonometric ratios	Computes exact algebraic values using special-angle formulas and identities.
`IdentityRewrite`	Reducible composite trigonometric expressions	Applies bounded identity rewrites (e.g., sum-to-product, double-angle formulas).
`SinusoidCanonicalize`	Linear combinations of sinusoidal terms	Rewrites expressions into canonical harmonic form $A\sin(\omega x+\varphi)+b$.
`PropertyDerive`	Canonical expressions with analytical property goals	Deduces period, amplitude, symmetry axes, extrema, and monotonicity intervals.
`EquationBaseSolve`	Trigonometric equations with active variable domains	Isolates elementary trigonometric functions and extracts all valid base roots in $[0, T)$.
`DomainRangeInequality`	Trigonometric inequalities or bounded expressions	Determines continuous domains, functional bounds, or base inequality intervals.
`PeriodicComplete`	Base solution units with target real domain $\mathbb{R}$	Lifts base-period roots or intervals to complete periodic solution sets over $\mathbb{R}$.
`AnswerValidate`	Candidate solution and target goal or candidate options	Verifies root substitution, checks periodic translation invariance, and matches options.

TABLE II: Solving Accuracy Comparison on the 795-Problem Benchmark.

Method	**Accuracy (%)**
Qwen3.5-Flash	63.4%
DeepSeek-V3	60.4%
Proposed Framework (Ours)	**74.8%**

C. Main Results

Table II presents the comparative solving performance on the complete benchmark. The proposed framework achieves an overall solving accuracy of 74.8%, successfully solving over 590 out of 795 problems. In comparison, both mainstream large language model baselines fail to achieve a 65.0% accuracy under identical problem statements.

This performance difference reflects the intrinsic difficulty of multi-step trigonometric reasoning. Although modern language models exhibit strong reasoning capabilities in general algebra, they frequently encounter difficulties when tracking strict angle conditions, verifying domain constraints, and generating complete periodic solution sets. In contrast, our framework coordinates domain-specific TMMs through structured reasoning strategies, effectively avoiding intermediate derivation errors and ensuring reliable problem solving.

VI. Discussion

The primary advantage of the proposed framework lies in the clear separation between domain-specific operation models and the high-level reasoning strategy. By encapsulating specialized mathematical rules into TMMs, the framework ensures that each intermediate derivation step strictly preserves domain constraints and angle conventions. Meanwhile, the DRS layer coordinates these models based on the evolving problem state rather than relying on rigid procedural routines. This decoupled architecture maintains mathematical soundness throughout the deduction process and allows new trigonometric rules to be integrated without altering the core reasoning mechanism.

Our work reveal several common misconceptions about trigonometric reasoning such as periodicity is merely a textual expression (e.g., "$+2k\pi$"). In fact, periodicity represents a defining characteristic of trigonometric functions that substantially affects the entire solving process. In practical problem solving, obtaining a complete solution set over the real domain requires identifying all valid solution branches and interval endpoints within the fundamental base period. Simply appending periodic notation to an incomplete base solution inevitably leads to missed roots or invalid intervals. Therefore, periodic completion must function as an integrated mathematical operation rather than a superficial formatting step, ensuring that local base solutions are reliably extended into globally complete solution sets.

In the context of intelligent education, generating transparent and tutorable derivation steps is as crucial as producing correct final answers. Because our framework records every intermediate state transition alongside its applicable mathematical model, the entire solving trajectory remains explicit and verifiable. These structured derivation paths can facilitate automated diagnosis of student errors and provide interpretable step-by-step feedback in intelligent tutoring systems.

Currently, our work focuses on text-based trigonometric function problems. In future research, we plan to extend this symbolic reasoning framework to multimodal trigonometric function problems involving geometric diagrams and functional graphs.

VII. Conclusion

This paper presents a meta-model-guided symbolic reasoning framework for solving trigonometric function problems in intelligent education systems. By formalizing specialized operations into structured trigonometric meta-models (TMMs), the framework ensures that intermediate state transitions strictly preserve angle conventions and domain constraints. Building upon these models, a decoupled reasoning strategy (DRS) is developed to dynamically schedule and execute TMMs based on evolving problem states and target solving goals. Experimental evaluations on a benchmark of 795 trigonometric function problems demonstrate that our framework achieves a solving accuracy of 74.8%, markedly outperforming large language model baselines. By generating mathematically verified and inspectable derivation trajectories, this work provides a reliable and scalable foundation for advancing automated mathematical problem solving.

Acknowledgment

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